

How to Build Tradeable Greeks: An Instruction Manual for Options Architecture

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Abstract

This manual explains how to turn the option Greeks from descriptive sensitivities into tradeable exposures. A Greek is not a separate security; it is a measurement of how an option portfolio should react when price, volatility, time, or rates change. To build a tradeable Greek, first name the desired exposure, then select instruments that naturally produce it, then hedge or limit the unwanted exposures that arrive with it. The manual walks through Delta, Gamma, Vega, Theta, and Rho as engineering problems. Each chapter states the economic target, lists practical building blocks, gives construction steps, explains what contaminating Greeks remain, and provides checks before execution. The emphasis is educational rather than predictive: the goal is not to recommend trades, but to teach the architecture of option positions. A long straddle, for example, is presented as a machine that manufactures long Gamma and long Vega while consuming Theta. An iron condor is presented as a machine that collects Theta while accepting short Gamma and short Vega. By the end, the reader should know how to design, size, hedge, monitor, and stress test portfolios whose central purpose is a chosen Greek exposure without confusing the exposure with the contract used to create it.

Core Operating Principle

A Greek is not normally something you buy from an exchange as a standalone asset.

A Greek is a sensitivity produced by a portfolio.

The task of building a “tradeable Greek” is therefore an engineering task:

Choose contracts, combine them into a structure, calculate the net sensitivity, hedge the unwanted sensitivities, and monitor how the structure changes as the market moves.

This manual treats each Greek as a target exposure and each option structure as a machine that manufactures that exposure.

Educational Use and Risk Note

This document is an instructional framework, not investment advice. Option structures can lose money quickly, and the risk profile shown by the Greeks is only a local approximation. The Greeks change when the underlying price, implied volatility, time to expiration, rates, dividends, liquidity, and market stress change.

Use the procedures here to understand construction logic, not to assume that any structure is safe, complete, or suitable for a particular account.

1 The Workshop Vocabulary

Before building tradeable Greeks, establish the language of the workshop.

The Instrument Is the Vehicle

The legal contract you trade may be:

- stock or an ETF;
- a future;
- a call option;
- a put option;
- a spread;
- a straddle or strangle;
- a calendar or diagonal spread;
- a butterfly or condor;
- a VIX option or VIX future;
- a variance swap, volatility swap, swaption, cap, floor, or other OTC derivative.

The Greek is the economic exposure produced by that vehicle.

The Greek Is the Output

For a portfolio with value V , underlying price S , implied volatility σ , time t , and interest rate r , a local approximation is:

$$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \nu d\sigma + \Theta dt + \rho dr.$$

Here ν denotes Vega. Some trading screens write Vega as *Vega* rather than ν , but the idea is the same.

This approximation says:

- Delta measures first-order exposure to the underlying price.
- Gamma measures how Delta changes when the underlying moves.
- Vega measures exposure to changes in implied volatility.
- Theta measures exposure to the passage of time.
- Rho measures exposure to interest-rate changes.

The approximation is local. It is useful near the current market state, but it does not guarantee the same behavior after large moves.

2 The General Construction Algorithm

Every tradeable Greek can be built with the same five-step recipe.

Step 1: Name the Target Exposure

Write the target in a complete sentence.

Examples:

- “I want positive Delta with limited downside.”
- “I want positive Gamma into an event.”
- “I want positive Vega while keeping Delta near zero.”
- “I want positive Theta, but I want maximum loss capped.”
- “I want exposure to rising rates through long-dated options.”

A vague sentence produces a vague structure. A precise sentence produces a buildable structure.

Step 2: Choose the Natural Greek Generator

Each Greek has instruments that naturally create it.

Target	Natural Generator	First Question
Positive Delta	Stock, futures, calls, short puts	Do I need convexity, or is the underlying cleaner?
Negative Delta	Short stock, short futures, puts, short calls	Do I need limited risk, or is the underlying cleaner?
Positive Gamma	Long options, straddles, strangles	Can the expected move pay for Theta?
Negative Gamma	Short options, short straddles, short strangles, condors	Am I being paid enough for jump risk?
Positive Vega	Long options, long straddles, long strangles, calendars, VIX products	Do I want volatility exposure or directional exposure?
Negative Vega	Short options, short straddles, short strangles, condors	Can I survive a volatility spike?
Positive Theta	Short premium structures, credit spreads, iron condors	What convexity or gap risk am I selling?
Negative Theta	Long premium structures, debit spreads, straddles	What move or volatility change justifies the decay cost?
Positive or negative Rho	Long-dated options, rate options, swaptions	Is rate exposure actually material?

Step 3: Identify the Contaminating Greeks

No common listed option structure creates one Greek in perfect isolation.

A long straddle may be built for Gamma, but it also brings Vega and negative Theta. An iron condor may be built for Theta, but it also brings short Gamma and short Vega. A long-dated call may be built for Delta, but it also brings Gamma, Vega, Theta, and Rho.

The construction task is to decide what to hedge, what to cap, and what to accept.

Step 4: Size the Position in Greek Units

Measure the position in the unit that matters.

For any Greek G , the portfolio exposure is:

$$G_{\text{net}} = \sum_i N_i \times M_i \times G_i,$$

where:

- N_i is the signed number of contracts or shares;
- M_i is the contract multiplier;
- G_i is the Greek of one unit of that instrument.

A buy order is not a risk plan. A Greek inventory is a risk plan.

Step 5: Stress Test Before Execution

Before placing the trade, ask how the position behaves if:

- the underlying rises sharply;
- the underlying falls sharply;
- implied volatility rises;
- implied volatility falls;
- one day passes;
- the position approaches expiration;
- liquidity disappears;
- the bid–ask spread widens;
- assignment or exercise becomes relevant;
- an event occurs before the hedge can be adjusted.

The Greek you want is only useful if the full structure can survive the scenarios you do not want.

3 The Builder’s Map of Common Structures

Use the following table as a map of the usual outputs produced by standard structures.

Structure	Delta	Gamma	Vega	Theta	Rho	Teaching Interpretation
Buy stock	+	0	0	0	small	Pure positive Delta.
Short stock	–	0	0	0	small	Pure negative Delta.

Structure	Delta	Gamma	Vega	Theta	Rho	Teaching Interpretation
Buy call	+	+	+	−	+	Direction plus convexity.
Sell call	−	−	−	+	−	Premium received for bearing upside risk.
Buy put	−	+	+	−	−	Downside exposure plus convexity.
Sell put	+	−	−	+	+	Premium received for bearing downside risk.
Long straddle	near 0	+	+	−	mixed	Movement and volatility machine.
Short straddle	near 0	−	−	+	mixed	Quiet-market premium machine.
Long strangle	near 0	+	+	−	mixed	Cheaper movement machine with wider breakevens.
Short strangle	near 0	−	−	+	mixed	Wider-range income machine with tail risk.
Long calendar	near 0	often − near strike	+	often + near strike	mixed	Term-structure and Vega machine.
Iron condor	near 0	−	−	+	small	Capped short-premium range machine.
Long butterfly	near 0	mixed	often −	often +	small	Pin-risk and range machine.

The signs in the table are starting points. Exact exposures depend on strike, maturity, money-ness, dividends, rates, and the volatility surface.

4 Build 1: Tradeable Delta

Objective

Build a position whose primary purpose is exposure to the direction of the underlying.

Positive Delta benefits from an upward move. Negative Delta benefits from a downward move.

Cleanest Instrument

The cleanest Delta instrument is usually the underlying itself: stock, ETF, or future.

If the only desired exposure is direction, the underlying is simpler than an option because it has no Gamma, Vega, or Theta.

When to Use Options Instead

Use options for Delta when you also want one of the following:

- limited premium risk;
- leverage;

- convex payoff;
- asymmetric upside or downside;
- a defined-risk spread;
- exposure that changes with the underlying.

Construction: Positive Delta

To build positive Delta:

1. Choose the underlying or bullish option structure.
2. Calculate the position Delta.
3. Decide whether the extra Greeks are useful or unwanted.
4. Cap risk if the structure can lose more than the planned amount.
5. Record the adjustment rule if Delta changes.

Common positive-Delta machines include:

- long stock;
- long futures;
- long calls;
- short puts;
- bull call spreads;
- bull put spreads.

Worked Construction

Suppose the target is +100 Delta.

A simple construction is:

Buy 100 shares.

A convex construction is:

Buy two call contracts with Delta +0.50 each and a contract multiplier of 100.

The approximate net Delta is:

$$2 \times 100 \times 0.50 = +100.$$

The stock construction is cleaner. The call construction adds positive Gamma, positive Vega, negative Theta, and usually positive Rho. That may be desirable if the trader wants upside convexity. It is undesirable if the trader wants only direction.

Construction: Negative Delta

To build negative Delta, use:

- short stock;
- short futures;
- long puts;
- short calls;
- bear put spreads;
- bear call spreads.

A long put gives negative Delta plus positive Gamma and positive Vega. A short call gives negative Delta plus negative Gamma and negative Vega. Those are very different machines even if the initial Delta looks similar.

Delta Checklist

Before executing a Delta trade, answer:

- Is the real goal direction only, or direction plus convexity?
- Is the Delta stable, or will Gamma change it rapidly?
- Is the option liquid enough to enter and exit efficiently?
- Is there event risk before the planned exit?
- Is the loss defined or open-ended?

5 Build 2: Tradeable Gamma

Objective

Build a position whose primary purpose is convexity.

Positive Gamma means the position becomes more long as the market rises and more short as the market falls.

Negative Gamma means the position becomes more short as the market rises and more long as the market falls.

Gamma is the Greek of acceleration. It controls how quickly Delta changes.

Construction: Positive Gamma

To build positive Gamma:

1. Buy options, preferably near the strike and maturity where Gamma is strongest.
2. If directional exposure is not desired, combine calls and puts or hedge with the underlying.
3. Estimate the daily Theta cost.
4. Ask whether expected realized movement can overcome that cost.
5. Define how often Delta will be rebalanced.

Common positive-Gamma machines include:

- long calls;
- long puts;
- long at-the-money straddles;
- long strangles;
- long short-dated options;
- long event options.

Long Gamma Straddle Recipe

A standard long-Gamma construction is:

Buy one at-the-money call and buy one at-the-money put with the same strike and expiration.

The initial Greek profile is approximately:

- Delta near zero;
- positive Gamma;
- positive Vega;
- negative Theta;
- mixed Rho.

This machine pays for convexity by spending time decay.

How to Make the Gamma Trade More Explicit

To make the trade more explicitly about Gamma:

1. Buy the straddle.
2. Calculate the net Delta.
3. Hedge the net Delta with stock or futures.
4. Recalculate Delta after meaningful price moves.
5. Rebalance according to a written rule, not emotion.

For example, if a call has Delta $+0.52$ and the put has Delta -0.48 , one straddle has net Delta $+0.04$ per share. With a 100-share multiplier, the position is $+4$ Delta. To start closer to Delta neutral, sell 4 shares or the equivalent futures exposure.

The Long-Gamma Economic Test

Long Gamma has a simple economic test:

Will realized movement plus any favorable volatility change exceed the Theta paid and the transaction costs?

If the underlying does not move, the structure decays. If implied volatility falls, the structure may lose even if the underlying moves modestly.

Construction: Negative Gamma

To build negative Gamma:

1. Sell options or option spreads.
2. Collect premium or Theta.
3. Cap the tail risk if the naked short option risk is unacceptable.
4. Stress test large moves.
5. Predefine adjustment or exit levels.

Common negative-Gamma machines include:

- short calls;
- short puts;
- short straddles;
- short strangles;
- iron condors;
- credit spreads.

Short Gamma Warning

Negative Gamma often feels comfortable while the market is quiet because Theta is positive. The danger is that losses can accelerate during large moves. A short-Gamma trader is selling convexity to someone else.

6 Build 3: Tradeable Vega

Objective

Build a position whose primary purpose is exposure to implied volatility.

Positive Vega benefits when implied volatility rises. Negative Vega benefits when implied volatility falls.

Vega is not the same as realized volatility. A position can be right about future realized movement and still lose if implied volatility changes unfavorably before exit.

Construction: Positive Vega

To build positive Vega:

1. Buy options or buy volatility-linked instruments.
2. Prefer maturities and strikes where Vega is meaningful.
3. Reduce unwanted Delta if the view is volatility rather than direction.
4. Understand the Theta cost.
5. Decide whether the trade needs Gamma, Vega, or both.

Common positive-Vega machines include:

- long calls;
- long puts;
- long straddles;
- long strangles;
- long calendars;
- long diagonals;
- longer-dated options;
- VIX options or futures;
- variance swaps or volatility swaps where available.

Long Vega Straddle Recipe

A simple positive-Vega construction is:

Buy an at-the-money call and an at-the-money put with the same expiration.

This creates long Vega, but it also creates long Gamma and short Theta. It is a good teaching structure because it shows the central tradeoff:

You own movement and volatility, but you pay for time.

Long Vega Calendar Recipe

A calendar spread can be used when the desired exposure is more about implied volatility and term structure than immediate movement.

A basic long calendar is:

Sell a near-term option and buy a longer-term option at the same strike.

Near the strike, the typical profile is:

- positive Vega;
- often positive Theta;
- often negative Gamma near the strike;
- Delta near zero at initiation if built at the money.

This is not the same machine as a long straddle. The straddle buys movement. The calendar buys longer-dated volatility while selling shorter-dated decay and short-term convexity.

Construction: Negative Vega

To build negative Vega:

1. Sell options or sell volatility-linked instruments.
2. Prefer defined-risk structures if volatility spikes could be dangerous.
3. Identify whether the short Vega is paired with short Gamma.
4. Stress test volatility expansion and price gaps at the same time.
5. Define the maximum acceptable loss before entering.

Common negative-Vega machines include:

- short calls;
- short puts;
- short straddles;
- short strangles;
- iron condors;
- short calendars;
- short volatility or variance products where available.

Vega Checklist

Before executing a Vega trade, answer:

- Am I trading implied volatility, realized volatility, or both?
- Is the position also strongly directional?
- Is the position long or short Gamma?
- How much Theta is being paid or collected?
- What happens if price gaps and implied volatility jumps together?

7 Build 4: Tradeable Theta

Objective

Build a position whose primary purpose is exposure to the passage of time.

Positive Theta benefits from time passing. Negative Theta is hurt by time passing.

In trader language, someone may say “I am long Theta” or “I am collecting Theta.” That usually means the trader has sold option premium.

Construction: Positive Theta

To build positive Theta:

1. Sell option premium.
2. Decide whether to use naked options or defined-risk spreads.
3. Identify the short Gamma and short Vega that usually accompany positive Theta.
4. Set maximum loss, adjustment, and exit rules.
5. Avoid confusing steady decay with guaranteed profit.

Common positive-Theta machines include:

- covered calls;
- cash-secured puts;
- credit spreads;
- short straddles;
- short strangles;
- iron condors;
- some butterflies;
- some calendars near the short strike.

Iron Condor Recipe

A classic defined-risk positive-Theta construction is the iron condor:

Sell an out-of-the-money put spread and sell an out-of-the-money call spread with the same expiration.

Its typical starting profile is:

- Delta near zero;
- negative Gamma;
- negative Vega;
- positive Theta;
- defined maximum loss if all legs remain in place.

The iron condor is a range machine. It is not free income. It collects premium in exchange for accepting losses if the underlying moves too far or volatility expands.

Construction: Negative Theta

To build negative Theta:

1. Buy option premium.
2. Define the expected movement or volatility expansion needed to justify the cost.
3. Decide whether the desired exposure is Delta, Gamma, Vega, or some combination.
4. Set a time stop if the thesis fails to develop.
5. Reassess as expiration approaches because Theta can accelerate.

Common negative-Theta machines include:

- long calls;
- long puts;
- debit spreads;
- long straddles;
- long strangles;
- long event-volatility structures.

Theta Checklist

Before executing a Theta trade, answer:

- What risk am I being paid to bear?
- Is the position short Gamma, short Vega, or both?
- Is maximum loss defined?
- What market move breaks the trade?
- What event could invalidate the assumption of quiet time decay?

8 Build 5: Tradeable Rho

Objective

Build a position whose primary purpose is exposure to interest rates.

Positive Rho generally benefits when rates rise. Negative Rho generally benefits when rates fall.

For many short-dated equity option trades, Rho is small compared with Delta, Gamma, Vega, and Theta. Rho becomes more important in long-dated options, interest-rate products, foreign-exchange options, bond options, swaptions, caps, floors, and structured products.

Construction: Positive Rho

To build positive Rho, consider:

- long-dated calls;
- short long-dated puts;
- rate options;
- payer swaptions;
- structures that benefit from higher discount rates or higher forward rates.

A long-dated equity call usually has positive Rho, but it also has Delta, Gamma, Vega, and Theta. If the intended trade is a rate view, those other Greeks may dominate unless they are hedged or deliberately accepted.

Construction: Negative Rho

To build negative Rho, consider:

- long-dated puts;
- short long-dated calls;
- receiver swaptions;
- structures that benefit from lower rates.

Rho Checklist

Before executing a Rho trade, answer:

- Is the option maturity long enough for Rho to matter?
- Is the trade really a rate trade or mostly a Delta/Vega trade?
- Does the product depend on dividends, funding, or foreign rates?
- Is liquidity sufficient for long-dated options?
- Would a direct rates instrument express the view more cleanly?

9 How to Isolate a Greek

The word “isolate” should be used carefully. In practice, traders usually reduce unwanted exposures rather than eliminate them perfectly.

Use the Underlying to Hedge Delta

If the desired exposure is Gamma or Vega rather than direction, hedge Delta with stock or futures. The hedge rule is:

$$H = -\Delta_{\text{portfolio}}.$$

If the option portfolio has +250 Delta, sell 250 shares or the equivalent futures exposure to begin near Delta neutral. If the portfolio later changes to −100 Delta, buy 100 shares or equivalent exposure to restore neutrality.

Delta hedging does not remove Gamma. It is required because Gamma keeps changing Delta.

Use Spreads to Cap Tail Risk

If a naked short option creates too much open-ended risk, convert it into a spread.

Examples:

- Sell a call and buy a higher-strike call to form a call credit spread.
- Sell a put and buy a lower-strike put to form a put credit spread.
- Combine both credit spreads to form an iron condor.

The purchased wings reduce premium received but define maximum loss.

Use Maturity to Shape Vega and Theta

Longer-dated options usually carry more Vega. Shorter-dated options usually have more intense Gamma and faster Theta near expiration.

This gives a construction rule:

Use strike to shape direction and convexity; use maturity to shape volatility and time exposure.

Calendars and diagonals are built from this idea. They are not merely cheaper options; they are term-structure machines.

Use Strike Selection to Shape Skew Exposure

Skew is not one of the five basic Greeks, but it matters when building tradeable Greeks. Out-of-the-money puts, at-the-money options, and out-of-the-money calls often carry different implied volatilities.

A trader who sells puts and buys calls is not only trading Delta. The trader is also trading the shape of the volatility surface.

10 Three Complete Build Sheets

This section shows how to turn a market opinion into a Greek architecture.

10.1 Build Sheet A: Long Vega, Near-Flat Delta

Market Sentence

“I believe implied volatility is too low, but I do not want the trade to depend mainly on direction.”

Candidate Machine

Buy an at-the-money straddle.

Construction Steps

1. Buy one at-the-money call.
2. Buy one at-the-money put.
3. Calculate the combined Delta.
4. Hedge residual Delta with stock or futures.
5. Record Vega, Gamma, and daily Theta.
6. Define the exit rule for volatility expansion, time decay, or adverse movement.

Expected Greek Profile

- Delta: near zero after hedge;
- Gamma: positive;
- Vega: positive;
- Theta: negative;
- Rho: usually mixed.

Main Lesson

A long straddle is not pure Vega. It is a long-volatility and long-convexity machine that must pay rent through Theta.

10.2 Build Sheet B: Positive Theta With Capped Loss

Market Sentence

“I believe the underlying will remain inside a range, and I want to collect time decay with defined risk.”

Candidate Machine

Sell an iron condor.

Construction Steps

1. Choose an expiration that matches the time horizon.
2. Sell an out-of-the-money put.
3. Buy a farther out-of-the-money put as protection.
4. Sell an out-of-the-money call.
5. Buy a farther out-of-the-money call as protection.
6. Confirm maximum profit, maximum loss, breakevens, and net Greeks.
7. Write the adjustment rule before entering.

Expected Greek Profile

- Delta: near zero at initiation if balanced;
- Gamma: negative;
- Vega: negative;
- Theta: positive;
- Rho: usually small or mixed.

Main Lesson

Positive Theta is compensation for bearing risk. In an iron condor, the risk is mostly movement beyond the range and volatility expansion.

10.3 Build Sheet C: Positive Gamma Into an Event

Market Sentence

“I expect a large move around an event, but I do not know the direction.”

Candidate Machine

Buy a short-dated straddle or strangle.

Construction Steps

1. Choose an expiration that captures the event.
2. Choose at-the-money strikes for stronger Gamma or out-of-the-money strikes for lower cost.
3. Calculate the premium paid and breakeven move.
4. Estimate the implied move embedded in the option prices.
5. Decide whether to hold through the event or exit before the event.
6. Plan for volatility crush after the event.

Expected Greek Profile

- Delta: near zero at initiation;
- Gamma: positive and often large;
- Vega: positive before the event;
- Theta: negative and often large;
- Rho: usually small.

Main Lesson

Long event Gamma can still lose money if the realized move is smaller than the implied move or if implied volatility collapses enough after the event.

11 The Greek Inventory Sheet

A trader building tradeable Greeks should keep a live inventory.

Line Item	Delta	Gamma	Vega	Theta	Rho
Instrument 1					
Instrument 2					
Instrument 3					
Instrument 4					
Hedge					
Net Exposure					
Target Exposure					
Difference					

Use the sheet this way:

1. Enter every leg with its signed quantity.
2. Multiply each Greek by contract quantity and multiplier.
3. Sum the columns.

4. Compare the net exposure with the target exposure.
5. Adjust the structure if the difference is too large.

The goal is not to make every unwanted Greek exactly zero. The goal is to know which risks remain and why they are acceptable.

12 The Monitoring Manual

A tradeable Greek is not built once. It must be maintained.

Daily Monitoring Questions

Ask these questions at each review:

- What is the current net Delta?
- Has Gamma made the Delta drift away from the target?
- Has implied volatility moved enough to change the Vega thesis?
- Is Theta helping or hurting at the expected rate?
- Has the position moved closer to expiration in a dangerous way?
- Have liquidity conditions changed?
- Is the original thesis still valid?

Adjustment Rules

Adjustments should be written before the trade is entered.

Examples:

- Rehedge Delta whenever net Delta exceeds a specified threshold.
- Exit a long straddle if implied volatility reaches a target level.
- Close a short premium trade if loss reaches a preset multiple of credit received.
- Roll a short option if the underlying reaches the short strike.
- Reduce size before an event if the position is not intended to carry event risk.

A trader without adjustment rules is not managing Greeks. The trader is reacting to price movement.

13 Common Construction Errors

Error 1: Confusing a Greek View With a Directional View

Buying a call because “volatility is cheap” may fail if the stock falls. If the view is volatility, the construction should reduce directional exposure.

Error 2: Calling Positive Theta Income

Theta is not salary. It is compensation for risk. The risk is often short Gamma, short Vega, gap risk, assignment risk, or liquidity risk.

Error 3: Ignoring the Volatility Surface

Different strikes and maturities can have different implied volatilities. Two options with the same expiration may not be equally cheap or expensive.

Error 4: Assuming Initial Greeks Stay Fixed

Greeks move. A position that begins Delta neutral may become strongly directional. A short-Gamma book can change quickly during large moves.

Error 5: Forgetting Transaction Costs

Gamma scalping and active Delta hedging require trading. Bid-ask spreads, commissions, slippage, and borrow costs can turn a theoretically good structure into a poor realized trade.

14 Final Build Checklist

Before executing any Greek architecture, complete this checklist.

Question	Required Answer
What Greek am I trying to build?	Name one primary target: Delta, Gamma, Vega, Theta, or Rho.
What sign do I want?	Positive or negative.
What instrument creates it naturally?	Identify the option, spread, underlying, future, or volatility product.
What unwanted Greeks come with it?	List each contaminating exposure.
Which unwanted Greeks will I hedge?	Specify the hedge instrument and hedge rule.
Which risks will I cap?	Use spreads, wings, position limits, or stop rules.
What is the maximum planned loss?	Write the number before entering.
What market scenario breaks the trade?	Define the move, volatility change, time decay, or rate shift.
When will I exit?	State profit target, loss limit, time stop, or thesis invalidation.
How will I monitor it?	Set review frequency and adjustment thresholds.

15 The Cleanest Mental Model

A tradeable Greek is not a floating Greek-letter asset. It is a designed sensitivity.

The construction sequence is:

Target exposure → choose instruments → calculate net Greeks → hedge contaminants
→ cap unacceptable risks → monitor and rebalance.

The Greek is the exposure.

The option architecture is the machine.

The risk process is the maintenance manual.

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